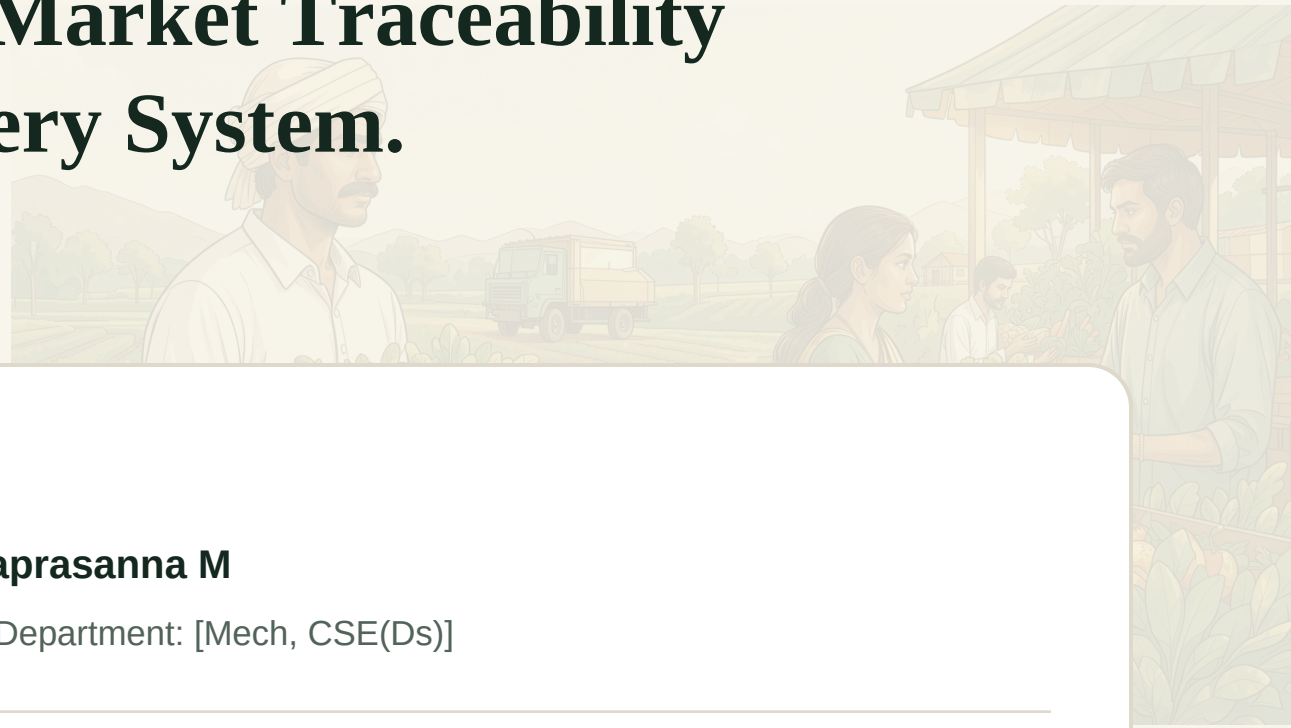
DOMAIN • AGRITRACE

Blockchain-Enabled Farm-to-Market Traceability and Transparent Price Discovery System.

From Farm to Market. Fair Prices. Trusted Produce.



TEAM 1

Presented by: Guru Prasanth S, Leun Hammi D, Jeyaprasanna M

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Supervisor: Dr. P. Chinnasamy | Designation: Associate Professor

PROJECT RECAP

THE NEED FOR AGRITRACE

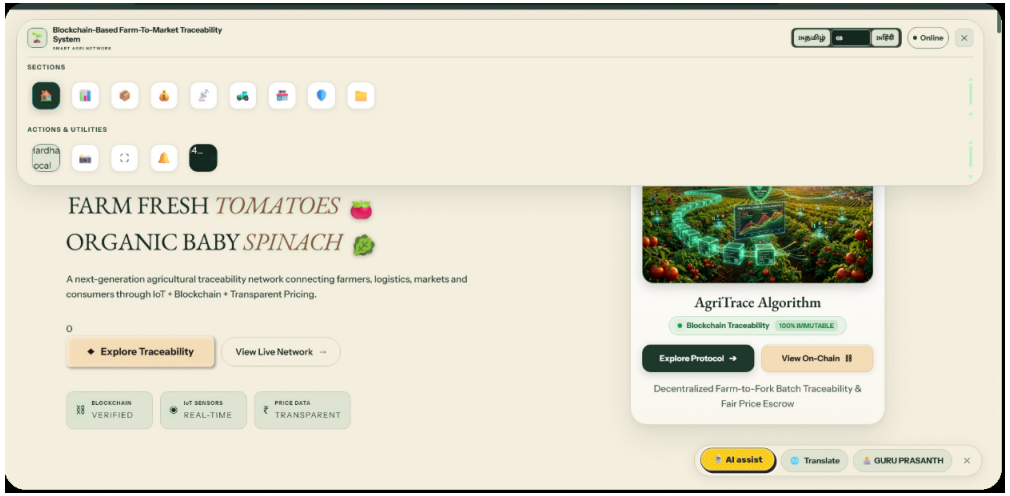
Existing challenges

- Multiple intermediaries can reduce direct buyer access.
- Market prices are not always easy to compare.
- Produce origin and handling details may be fragmented.
- Agricultural information is spread across platforms.



AgriTrace solution

- **Digital farmer-to-buyer marketplace with separate portals.**
- **Transparent crop-price info and district-wise comparison.**
- **Government datasets, crop information, & blockchain trace.**
- **QR-based verification where supported.**



Farmer → AgriTrace platform → Buyer | Price transparency • Digital marketplace • Traceability

https://guru-prasanths.github.io/AgriTrace-/

PRESENTATION ROUTE

AGENDA – EXPLORING THE AGRITRACE ECOSYSTEM

01
Project overview and objectives

02
Farmer and seller authentication

03
Digital agricultural marketplace

04
Crop listings and order management

05
Tamil Nadu agriculture datasets

06
District-wise market agricultural price comparison

07
Blockchain-based and QR traceability verification

08
AgriTrace Points and accessibility

09
Technology architecture and future scope

FUNCTIONAL ARCHITECTURE

AGRITRACE – SYSTEM WORKFLOW AND FUNCTIONAL ARCHITECTURE

Authentication

Farmer & Seller portals · email/password · Google auth · role-based access

Listing

Crop name · images · quantity & unit · price & location · quality & availability

Marketplace

Browse · compare listings · submit requests or offers · connect with farmers

Orders

Track requests · manage status · coordinate interactions · record transactions

Price discovery

Tamil Nadu datasets & official resources · tomato compare Salem, Madurai, CBE

Traceability

Produce Batch → Digital Record → Blockchain Tx → QR Verification

AgriTrace Points

Eligible activities · completion rewards · daily eligibility · transaction history

Accessibility

Simple navigation · responsive interface · readable text · voice assistant

ARCHITECTURE: HTML5 | CSS3 | JS → Node.js | Express.js → PostgreSQL | Solidity | Hardhat | MetaMask | ethers.js → TN Datasets

- Comparison fields: Minimum, maximum, modal price, market, date, unit. Tomato values shown are illustrative placeholders.
- Traceability boundary: supported records only, not a deployed production blockchain. QR verification where supported.

CLOSING

THANK YOU

